Primary-hit spacing and MC graph fragmentation

28 October 2026

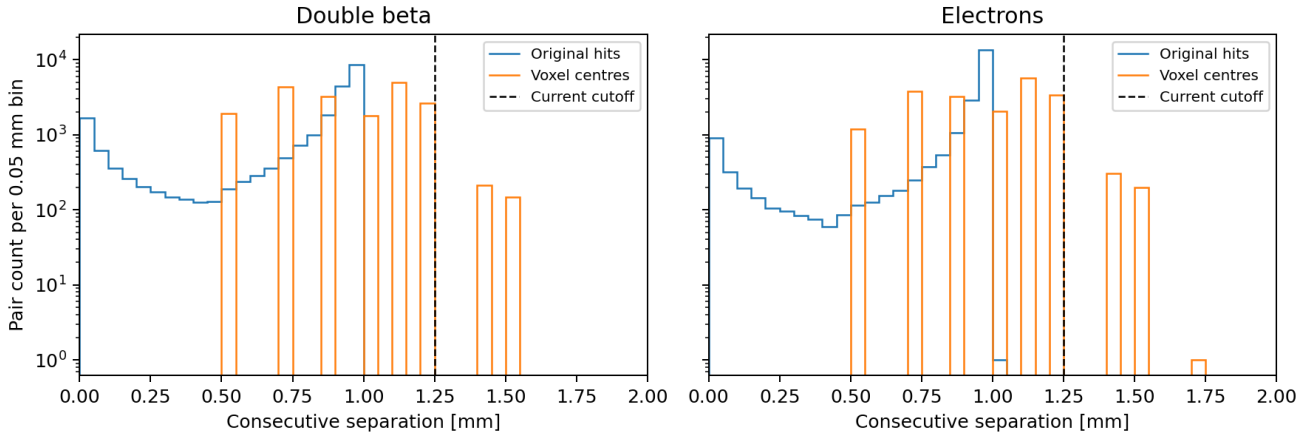
The first 100 catalogue events per sample are used. Each actual primary electron is ordered separately by MC time and hit ID. Consecutive stored-hit distances are spatial chords, not G4 path lengths. The approximately 1 mm G4 step is a user-provided simulation input. No diffusion, energy threshold or fiducial cut is applied. Coordinates and distances use Float32 mm.

Voxel centres follow the viewer convention: each coordinate is replaced by $(\lfloor x/0.5 \rfloor + 0.5)0.5$ mm. Voxel-transition statistics exclude successive hits mapped to the same cell. Revisits remain in trajectory order.

Sample	Positions	Pairs	Median [mm]	P99 [mm]	Max [mm]	Above 1.25 mm [%]
Double beta	Original hits	21880	0.930	0.998	1.000	0.00
Double beta	Voxel centres	19187	1.000	1.414	1.500	1.86
Electrons	Original hits	21273	0.970	0.999	1.003	0.00
Electrons	Voxel centres	19834	1.000	1.414	1.732	2.53

Double beta: 200 primary electrons, 0 time/hit-ID ordering disagreements, 0 equal-time pairs. 357 pairs with original separation ≤ 1.05 mm acquire a voxel separation > 1.25 mm.

Electrons: 100 primary electrons, 0 time/hit-ID ordering disagreements, 0 equal-time pairs. 501 pairs with original separation ≤ 1.05 mm acquire a voxel separation > 1.25 mm.



Every original primary-hit link in this scan passes the 1.25 mm cut. Voxel-centre quantisation moves some of these links beyond the cut. This directly demonstrates a source of fragmentation of a continuous primary trajectory. All observed voxel-transition links pass a 2 mm cut. The 2 mm radius is therefore supported for further inspection with 0.5 mm MC voxels; this is not a measurement of background rejection or a full-sample selection validation. Threshold removal and secondary deposits are separate effects. Viewer settings remain unchanged.

Numerical source: `results/event_analysis/mc_graph_diagnostic/hit_spacing.h5`.